Paper 2

1 **(a)** **(i)** atmospheric pressure = 9.10 × 10^4 Pa A1

(ii) (9.15 – 9.10) × 10^4 = ρ_m × 9.81 × (0.17 – 0.10) C1

ρ_m = 728 kg m^{-3} A1

(b) **(i)** pressure at top surface of cube = 9.135 × 10^4 Pa (from graph)

pressure at bottom surface of cube = 9.180 × 10^4 Pa (from graph) C1

Upthrust = (9.180 – 9.135) × 10^4 × $(0.051)^2$ C1

= 1.17 N A1

(ii) force = 4 – 1.17 = 2.83 N A1

(c) Remove the cube and check if spring returns to original length B1

(d) **(i)** free body diagram of upper ball, three forces: 1. weight, 2. horizontal force by wall on ball and 3. force by lower ball on upper ball.

Angle is 45° between horizontal and the dotted line.

So tan(45°) = (horizontal force) / (weight) C1

Horizontal force = weight = 1.67 N A1

OR taking moment about axis through point of contact between the balls:

Same moment arm C1

Hence *F* = weight of ball = 1.67 N A1

(ii) $F = \sqrt{(1.67^2 + 1.67^2)}$ C1

= 2.36 N A1

2 **(a)** g = (6.1 ± 0.1) N m^{-1} C1